

GRADUATION PROJECT

PROGRESS REPORT

Ai-Powered Triage System

Medical RAG System for
Condition Diagnosis

Using Retrieval-Augmented Generation

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WEEKLY PROGRESS REPORT

AI-Powered Medical Triage System (RAG-Based CRM System)
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1. Executive Summary

During this reporting period, the team focused on establishing the foundational infrastructure required for the system. The main achievements include the completion of the database design and implementation, along with initiating the evaluation phase of the Retrieval-Augmented Generation (RAG) model.

This phase marks a critical transition from planning to execution, where the system begins evolving into a fully functional AI-powered CRM medical triage platform.

2. Detailed Progress

2.1 Database Design and Implementation

The team successfully completed the full database structure, which serves as the backbone of the entire system.

Key Work Done:

Designed a structured database to store:

- Medical conditions and datasets
- Symptoms and diagnostic mappings
- User queries and system responses
- CRM-related data (patients, interactions, logs)

Ensured the database supports:

- Fast retrieval for RAG processing
- Scalability for future expansion
- Clean integration with backend modules

Technical Considerations:

- Logical schema design to align with RAG architecture
- Data normalization to reduce redundancy
- Preparation for integration with APIs and system interfaces

Outcome:

Database is now fully completed and ready for system integration

2.2 System Development Planning

A structured development roadmap was defined to guide the next phase of the project.

Planned System Transformation:

- Managing user/patient interactions
- Tracking diagnostic sessions
- Providing intelligent AI-based recommendations

Planned Modules:

- User Management Module
- Diagnosis & RAG Processing Module
- CRM Interaction & Logging Module
- Evaluation & Feedback Module

Outcome:

Clear development direction and modular architecture established

2.3 RAG Model Evaluation (Initiation Phase)

Following database completion, the team began the evaluation of the RAG model.

Objectives of Evaluation:

- Measure accuracy of retrieved medical conditions
- Analyze relevance of responses to user symptoms
- Evaluate system response time and efficiency
- Assess reasoning quality of generated outputs

Evaluation Focus Areas:

- Retrieval quality
- Ranking effectiveness
- Reasoning mechanism improvement
- Output clarity and usability

Initial Observations:

- Better reasoning refinement needed

- Improved ranking accuracy required
- Fine-tuning of response generation

Next Actions:

- Iterative testing
- Adjusting reasoning logic
- Enhancing model performance

3. Key Achievements

- ✓ Successfully completed **database design and implementation (100%)**
- ✓ Established **clear system development roadmap**
- ✓ Initiated **RAG model evaluation phase**
- ✓ Defined system transition toward a **CRM-based AI platform**

4. Current Project Status

Component	Status	Completion
Database	Completed	100%
System Architecture	Defined	80%
CRM Development	Not Started	0%
RAG Evaluation	In Progress	20%

5. Challenges and Considerations

5.1 System Integration Complexity

- Ensuring smooth communication between:
 - Database
 - RAG system
 - CRM modules

5.2 RAG Model Optimization

- Need to improve:
 - Accuracy of retrieved results
 - Logical reasoning in generated responses

5.3 Performance and Scalability

- Preparing the system to handle:
 - Larger datasets
 - More concurrent users

6. Next Steps (Upcoming Week)

6.1 Development Phase

- Begin implementation of:
 - CRM system modules
 - Backend logic for system integration

6.2 RAG Optimization

- Continue evaluation and testing
- Improve reasoning algorithm
- Enhance retrieval accuracy

6.3 System Integration

- Connect database with:
 - RAG model
 - Application backend

6.4 Testing

- Start initial system-level testing
- Validate workflow from input → diagnosis → output

7. Conclusion

This reporting period represents a significant step forward in the development of the AI-Powered Medical Triage System. The successful completion of the database establishes a robust and scalable foundation for the system, ensuring efficient data management and seamless integration with upcoming modules.

In addition, the initiation of the RAG model evaluation phase marks the transition toward performance optimization and intelligent decision-making enhancement. Early observations indicate that while the system is functional, further refinement is required to improve retrieval accuracy, reasoning quality, and overall response effectiveness.

Overall, the project is progressing in alignment with its planned objectives. The upcoming phase will focus on system development, integration of CRM functionalities, and continuous improvement of the AI model. These efforts will play a critical role in delivering a reliable, efficient, and user-centered medical triage solution.